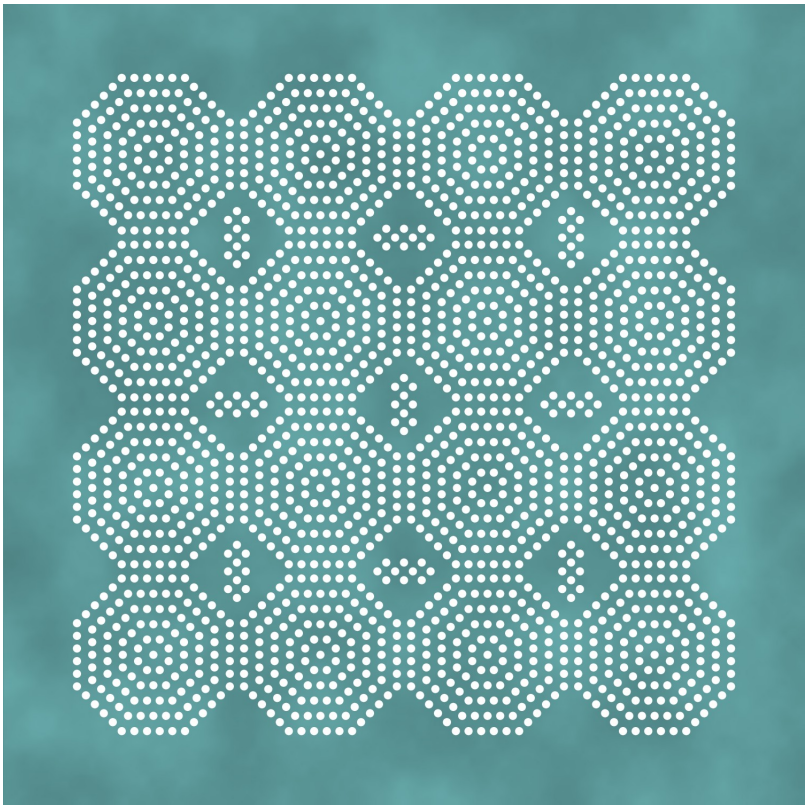
Picture for 2026



Centered Octogonal Numbers

$$
\begin{aligned}
O_1 &= 1 \\
O_2 &= 1 + 8 = 9 \\
O_3 &= 1 + 8 + 16 = 25 \\
&\vdots \\
O_n &= 1 + 8 + 16 + \cdots + 8\,(n-1) \\
&= 1 + 8\,(1 + 2 + \cdots + (n-1)) \\
&= 1 + 8\left(\frac{(n-1)\,(n)}{2}\right) \\
&= 1 + 4\,(n-1)\,(n) \\
&= 1 + 4n^2 - 4n
\end{aligned}
$$

Now, $O_6 = 121$, and $16\,(121) + 9\,(10) = 2026$.

Also, $2026 = 2\,(1013)$, so 2026 is the product of two primes.

Cantor Set

Geometrically, start with the unit interval.

$$C_0 = [0, 1] = \{x \mid 0 \leq x \leq 1\}$$

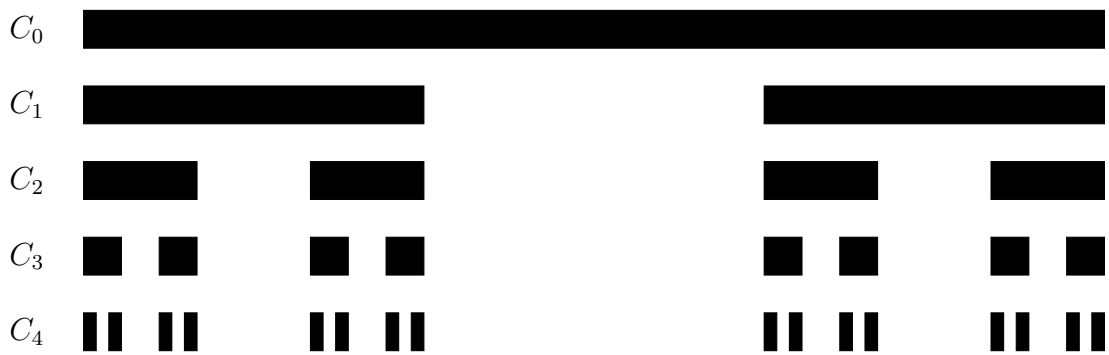
Rule for iteration: Remove the middle third

$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] = \left\{x \mid 0 \leq x \leq \frac{1}{3} \text{ or } \frac{2}{3} \leq x \leq 1\right\}$$

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]$$

$$C_3 = \left[0, \frac{1}{27}\right] \cup \left[\frac{2}{27}, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{7}{27}\right] \cup \left[\frac{8}{27}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{19}{27}\right] \cup \left[\frac{20}{27}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, \frac{25}{27}\right] \cup \left[\frac{26}{27}, 1\right]$$

Graphically, this looks like:



Repeat this process ad infinitum.

C_n consists of 2^{n-1} closed intervals of length $\left(\frac{1}{3}\right)^{n-1}$.

Cantor Set: $C = \bigcap_{n=1}^{\infty} C_n$, and has total length 0.

Base 3 (Ternary Numbers)

- The possible digits are 0, 1, 2

Decimal	Ternary
0	0
1	1
2	2
3	10
4	11
5	12
6	20
7	21
8	22
9	100
10	101
1/3	0.1
2/3	0.2
1/9	0.01
2/9	0.02
7/9	0.21
8/9	0.22

The Cantor Set consists of all points of the form:

$$0.c_1c_2c_3\ldots \qquad \text{where } c_i \in \{0,2\}$$

What about $\left(\frac{1}{3}\right)_{10} = 0.1_3$?

Consider $0.0\overline{222}_3$, and convert it to decimal

$$0\left(\frac{1}{3}\right) + 2\left(\frac{1}{9}\right) + 2\left(\frac{1}{27}\right) + \cdots = \frac{2}{9}\left(1 + \frac{1}{3} + \frac{1}{3^2} + \ldots\right) = \frac{2}{9}\left(\frac{1}{1-\frac{1}{3}}\right) = \frac{1}{3}$$

The idea is removing middle thirds if equivalent to saying that we are keeping only the numbers that have digits that are only 0's or 2's.

It can be helpful to have different ways to represent a set depending on what we are trying to do with it.

Claim: The Cantor Set C is uncountable.

Proof. Suppose the Cantor Set is countable. Now, list all members of the set:

$$a_1 = 0.a_{11}a_{12}\ldots$$

$$a_2 = 0.a_{21}a_{22}\ldots$$

Where each $a_{ij} \in \{0,2\}$.

Now, let $b = 0.b_1b_2\ldots$ where $b_i = \begin{cases} 0 & \text{if } a_{ii} = 2 \\ 1 & \text{if } a_{ii} = 0 \end{cases}$.

This is an element of C that is not on the list, thus C is not countable. $\square$